Recombinant expression by ZY auto-induction

What you're going to get / What you expect

Overexpression of the target enzyme in E. coli starting from a colony on the selective plate, without IPTG or OD monitoring: the ZY medium induces the T7 system automatically as the culture reaches high density.

Materials list (per 10 mL production culture)

- Selective plate with colonies of the transformed (DE3) expression strain.
- LB medium with the relevant antibiotic (for the pre-inoculum).
- Supplemented ZY auto-induction medium with the relevant antibiotic.
- Orbital shaker at 37 ºC and at 20 ºC.

Steps

1. Pick a single colony from the selective plate into 5 mL of LB with antibiotic and grow overnight at 37 ºC under orbital agitation (pre-inoculum).
2. Add the pre-inoculum directly to 10 mL of supplemented ZY auto-induction medium.
3. Incubate overnight at 20 ºC. The glucose in the medium is consumed first and the lactose then induces the system automatically, so no OD600 monitoring or reagent addition is needed.
4. Collect biomass for the expression test.

Resources & Downloads

- ZY auto-induction medium: Studier F.W. (2005), Protein Expression and Purification 41:207-234.

Errores comunes / troubleshooting

- Lower overall yield than IPTG for several of these constructs: prefer IPTG where the highest expression is needed.
- Protein made but insoluble: apply the solubilisation strategies (30 ºC induction, chaperones, oxidising or rare-codon strains).